

MIT OpenCourseWare

7.05

General Biochemistry 7.05 — General Biochemistry May 2026 **EXTREMELY****HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (4 points) Analyze the limitations of standard approaches to Protein Structure when applied to edge cases in 7.05. Construct explicit counterexamples showing where intuitive reasoning fails.
2. (3 points) Construct an original proof-level problem involving Enzyme Kinetics for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
3. (4 points) Prove the fundamental theorem concerning Metabolism in the context of 7.05 (General Biochemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
4. (5 points) Prove the fundamental theorem concerning Signal Transduction in the context of 7.05 (General Biochemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
5. (4 points) Analyze the limitations of standard approaches to Bioenergetics when applied to edge cases in 7.05. Construct explicit counterexamples showing where intuitive reasoning fails.
6. (4 points) Construct an original proof-level problem involving Protein Structure for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
7. (5 points) Construct an original proof-level problem involving Enzyme Kinetics for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
8. (9 points) Analyze the limitations of standard approaches to Metabolism when applied to edge cases in 7.05. Construct explicit counterexamples showing where intuitive reasoning fails.

9. (2 points) Construct an original proof-level problem involving Signal Transduction for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
10. (4 points) Prove the fundamental theorem concerning Bioenergetics in the context of 7.05 (General Biochemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
11. (3 points) Prove the fundamental theorem concerning Protein Structure in the context of 7.05 (General Biochemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (2 points) Prove the fundamental theorem concerning Enzyme Kinetics in the context of 7.05 (General Biochemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
13. (3 points) Analyze the limitations of standard approaches to Metabolism when applied to edge cases in 7.05. Construct explicit counterexamples showing where intuitive reasoning fails.
14. (5 points) Prove the fundamental theorem concerning Signal Transduction in the context of 7.05 (General Biochemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
15. (6 points) Construct an original proof-level problem involving Bioenergetics for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
16. (2 points) Construct an original proof-level problem involving Protein Structure for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
17. (7 points) Prove the fundamental theorem concerning Enzyme Kinetics in the context of 7.05 (General Biochemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
18. (3 points) Analyze the limitations of standard approaches to Metabolism when applied to edge cases in 7.05. Construct explicit counterexamples showing where intuitive reasoning fails.
19. (4 points) Analyze the limitations of standard approaches to Signal Transduction when applied to edge cases in 7.05. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (4 points) Analyze the limitations of standard approaches to Bioenergetics when applied to edge cases in 7.05. Construct explicit counterexamples showing where intuitive reasoning fails.
21. (2 points) Construct an original proof-level problem involving Protein Structure for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (4 points) Prove the fundamental theorem concerning Enzyme Kinetics in the context of 7.05 (General Biochemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (2 points) Analyze the limitations of standard approaches to Metabolism when applied to edge cases in 7.05. Construct explicit counterexamples showing where intuitive reasoning fails.
24. (5 points) Construct an original proof-level problem involving Signal Transduction for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
25. (4 points) Construct an original proof-level problem involving Bioenergetics for 7.05 (General Biochemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

Solutions

Solution 1:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Protein Structure. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 2:

Solution approach: First recall the definitions and key theorems related to Enzyme Kinetics from 7.05. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 3:

The proof follows from the definitions and properties established in 7.05. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 4:

The proof follows from the definitions and properties established in 7.05. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 5:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Bioenergetics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 6:

Solution approach: First recall the definitions and key theorems related to Protein Structure from 7.05. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 7:

Solution approach: First recall the definitions and key theorems related to Enzyme Kinetics from 7.05. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 8:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Metabolism. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 9:

Solution approach: First recall the definitions and key theorems related to Signal Transduction from 7.05. Then apply the appropriate techniques to construct a rigorous

argument. The solution must be self-contained and reference only the material from the course.

Solution 10:

The proof follows from the definitions and properties established in 7.05. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 11:

The proof follows from the definitions and properties established in 7.05. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:

The proof follows from the definitions and properties established in 7.05. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 13:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Metabolism. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 14:

The proof follows from the definitions and properties established in 7.05. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 15:

Solution approach: First recall the definitions and key theorems related to Bioenergetics from 7.05. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 16:

Solution approach: First recall the definitions and key theorems related to Protein Structure from 7.05. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 17:

The proof follows from the definitions and properties established in 7.05. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 18:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Metabolism. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 19:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Signal Transduction. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Bioenergetics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 21:

Solution approach: First recall the definitions and key theorems related to Protein Structure from 7.05. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 22:

The proof follows from the definitions and properties established in 7.05. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 23:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Metabolism. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 24:

Solution approach: First recall the definitions and key theorems related to Signal Transduction from 7.05. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 25:

Solution approach: First recall the definitions and key theorems related to Bioenergetics from 7.05. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.